Administrivia

- Activity 2 – Thursday Sept 18
  - Bring your laptop

- HW 1 is out
  - Due in 10 days

CompSci 469/469G
Intro to Computer Security
Set-UID Programs, Environment Variables

Source

- Computer Security-A Hands on Approach (3rd Edition), Wenliang Du.
  - Referenced Chapter: 2, 3

Set-UID Programs

RUID = the UID of the user who started it
EUID = the UID of the file's owner

- When Bob runs ./mycat (not privileged), the kernel sets:
  - RUID = Bob's UID
  - EUID = Bob's UID
  - RUID = EUID

- When a program become privileged with set-uid whenever anyone runs this file, it runs with file owner's ID.
  - RUID = Bob's UID
  - EUID = Root's UID (owner of ./mycat)

Password Dilemma

- Changing password means adding an entry to /etc/shadow
  - /etc/shadow contains hashed passwords not plain text
- Permissions of /etc/shadow File:

```
ls -l /etc/shadow
-rw-r----- 1 root shadow 723 Sep 11 20:05 /etc/shadow
```

- How would normal users change their password?
  - Passwd command invoke /bin/passwd program [a compiled binary program]
  - Permissions of /bin/passwd File:

```
ls -l /bin/passwd
-rwsr-xr-x 1 root root 72056 May 30  2024 /bin/passwd
```



Demo: Turn a Program into Set-UID Program

```
cp /bin/cat ./mycat
bob@52b7cff8b811:~$ ls -l mycat
-rwxr-xr-x 1 bob stu 67928 Sep 11 20:40 mycat

bob@52b7cff8b811:~$ ./mycat /etc/shadow
./mycat: /etc/shadow: Permission denied
```

Making a copy of cat program to be safe

mycat is not a privileged program

Bob can't access shadow file

Demo: Turn a Program into Set-UID Program

```
cp /bin/cat ./mycat
bob@52b7cff8b811:~$ ls -l mycat
-rwxr-xr-x 1 bob stu 67928 Sep 11 20:40 mycat

bob@52b7cff8b811:~$ ./mycat /etc/shadow
./mycat: /etc/shadow: Permission denied
```

```
bob@52b7cff8b811:~$ sudo chown root ./mycat
bob@52b7cff8b811:~$ sudo chmod 4755 ./mycat
```

Chown command: change ownership of a file

To set set-uid bit

Demo: Turn a Program into Set-UID Program

```
cp /bin/cat ./mycat
bob@52b7cff8b811:~$ ls -l mycat
-rwxr-xr-x 1 bob stu 67928 Sep 11 20:40 mycat

bob@52b7cff8b811:~$ ./mycat /etc/shadow
./mycat: /etc/shadow: Permission denied
```

```
bob@52b7cff8b811:~$ sudo chown root ./mycat
bob@52b7cff8b811:~$ sudo chmod 4755 ./mycat

bob@52b7cff8b811:~$ ls -l ./mycat
-rwsr-xr-x 1 root stu 67928 Sep 11 20:40 ./mycat
```

Chown command: change ownership of a file

To set set-uid bit

mycat Become a privileged program

Demo: Turn a Program into Set-UID Program

```
cp /bin/cat ./mycat
bob@52b7cff8b811:~$ ls -l mycat
-rwxr-xr-x 1 bob stu 67928 Sep 11 20:40 mycat

bob@52b7cff8b811:~$ ./mycat /etc/shadow
./mycat: /etc/shadow: Permission denied
```

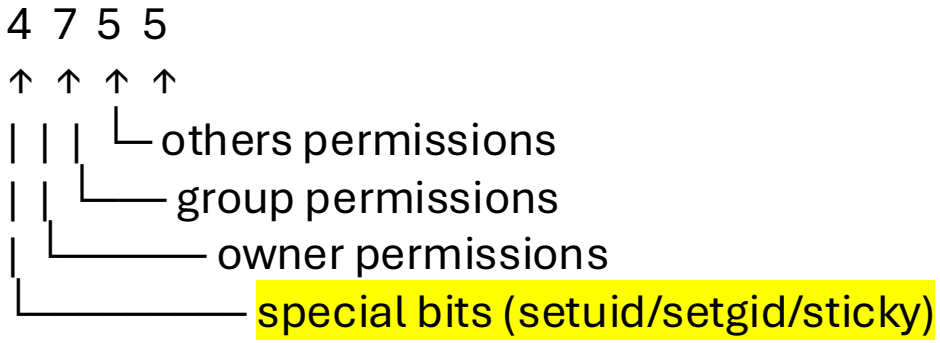
```
bob@52b7cff8b811:~$ sudo chown root ./mycat
bob@52b7cff8b811:~$ sudo chmod 4755 ./mycat
bob@52b7cff8b811:~$ ls -l ./mycat
-rwsr-xr-x 1 root stu 67928 Sep 11 20:40 ./mycat

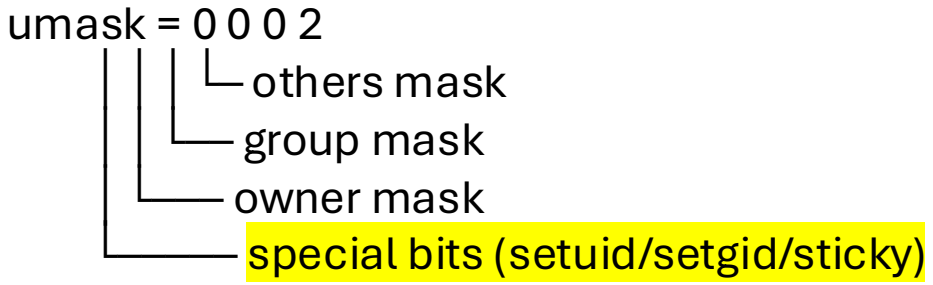
bob@52b7cff8b811:~$ ./mycat /etc/shadow
root:*:20305:0:99999:7:::
daemon:*:20305:0:99999:7:::
bin:*:20305:0:99999:7:::
```

Now Bob can access shadow file.
We didn't change the file's permissions;
instead, we made the program that accesses it run with elevated privileges.

Breaking Down 4755

chmod 4755 ./mycat





Leftmost digit - Special bits

4 → 100 setuid bit set - process runs with EUID = file owner's UID.

2 → 010 - setgid bit set – process runs with EGID = file's group ID.

1 → 001 - sticky bit set - Prevents users from deleting files they don't own, even if they have write access.

Special bits

setuid bit - process runs with EUID = file owner's UID.

setgid bit – process runs with EGID = file's group ID.

sticky bit - Prevents users from deleting files they don't own, even if they have write access.

To set setuid bit – chmod u+s ./mycat

To set setgid bit - chmod g+s ./mycat

To set sticky bit – chmod +t ./src

Changing owner of a Privileged program

Linux clears the setuid bit whenever the owner of a file is changed with chown.

```
ls -l ./mycat
-rwsr-xr-x 1 root bob 67928 Sep 11 20:40 ./mycat

sudo chown bob ./mycat
ls -l mycat
-rwxr-xr-x 1 bob bob 67928 Sep 11 21:28 mycat
```

Why? – to prevent privilege escalation attack

- If bob could chown it to themselves without losing the s bit, they would instantly have a setuid-cat program that runs with their privileges.

Environment Variable

What happens when you run ls?

- On Linux, the ls program is in usr/bin/ls.
- To see the path of the program: which ls
- When you type ls, the shell doesn't magically know what ls is.
- It has to find the program somewhere on disk.

How does the shell find it?

- The shell checks the PATH variable (a list of directories).
- It searches them in order until it finds a matching program.

```
echo $PATH
/usr/local/bin:/usr/bin:/bin:/usr/local/sbin:/usr/sbin:/sbin
```

Environment Variable

- PATH is just one example of an environment variable.
- Environment variables are values passed to processes by the shell.
- You can see them all with: env

- Examples:
  - PATH → where to find programs.
  - HOME → your home directory.
  - USER → your username.

Attacks via Environment Variables

- Attacker can manipulate the PATH variable and control how actual program works
- export command in Linux is used to set and export environment variables

```
export PATH=.:$PATH
```

- For example, cal is a system command to see the calendar. Attacker can write malicious program named as cal and change path variable to execute it

```
bob@52b7cff8b811:~$ cal
   September 2025
Su Mo Tu We Th Fr Sa
    1  2  3  4  5  6
 7  8  9 10 11 12 13
14 15 16 17 18 19 20
21 22 23 24 25 26 27
28 29 30
```

Attacks via Environment Variables

How can attacker modify the PATH environment variable to attack the system?

```
bob@52b7cff8b811:~$ echo 'echo "SYSTEM COMPROMISED"' > cal
bob@52b7cff8b811:~$ chmod +x cal
bob@52b7cff8b811:~$ export PATH=.:$PATH
bob@52b7cff8b811:~$ echo $PATH
.:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/bin
bob@52b7cff8b811:~$ cal
SYSTEM COMPROMISED
```